

Agentic evaluation of function prediction tools yields qualitative insights into systematic errors

Abstract

New protein-function predictors appear faster than annotation databases can assess them manually. Aggregate benchmarks measure agreement with annotation snapshots, but they do not directly evaluate the free-text functional summaries and rationales emitted by newer reasoner-style systems or expose biologically patterned errors hidden by averages.

We describe AI-AUGR (Assessment via Unified Gene-evidence Review), an agentic curation pipeline that synthesizes literature, domain architecture, and existing annotations to evaluate predictions gene by gene, producing structured, human-readable qualitative assessments.

We apply AI-AUGR to BioReason-Pro using two linked cohorts. ARGO139 is a 139-export collected cohort for RL narrative review; the performance set excludes one wrong-input case (n=138) and flags seven sequence-truncated inputs. ARGO95 is the 95-gene subset with 955 HuggingFace-catalogue SFT GO-term predictions. The local AIGR references are agent-adjudicated and have mixed review maturity; they are not independently expert-signed ground truth.

On the ARGO139 performance set, mean RL narrative correctness was 4.0/5 and completeness was 2.9/5. A blinded 20-gene second review yielded quadratic-weighted kappa 0.950 for correctness and 0.744 for completeness. In ARGO95, 679/955 SFT terms were correct but non-novel (632 exact frozen-GOA matches and 47 supported by other established local evidence), 151/955 were classified as NPI, PLI, or REP, and 24/955 were correct known-literature functions absent from the frozen GOA snapshot. The review identifies recurring errors involving pseudoenzymes, localization, paralog identity, organism context, and generated database-style prose, showing how agentic biological-validity review can complement rather than replace aggregate and expert evaluation.

1 Introduction

Functional annotation of gene products is one of the central tasks of modern molecular biology, underpinning high-throughput analyses of molecular data. The Gene Ontology (GO) project organizes these annotations, and coordinated both manual curation efforts and the assessment and application of annotation pipelines[1].

Annotations in GO can be classified in two categories: **(i) literature-based expert curation**, in which expert curators and encode findings from papers as structured, evidence-coded annotations; and **(ii) computational inference**, which propagates function from literature-based knowledge across the tree of life.

Traditional computational pipelines such as InterPro2GO, PANTHER/PAINT, and orthology transfer are attractive because their inferences are transparent: annotations can be traced back to a family, domain signature, phylogenetic node, or orthology assertion. Over-prediction can usually be easily corrected, for example by weakening a mapping between a domain/family and a GO term. Newer machine-learning predictors and protein language-model systems are less rule-like. They may consume sequence, structure, organism context, or model-derived embeddings directly, and some now emit free-text rationales or functional summaries alongside term lists. This makes evaluation harder: the prediction is no longer only a set of GO terms, and the provenance chain is often less explicit.

This presents a challenge for assessing whether predictions should be included alongside trusted methods in the main annotation corpus. The Critical Assessment of Functional Annotation (CAFA) project [2–4] evaluates function annotation pipelines, using temporal holdout against GO database snapshots and reports aggregate agreement scores.

CAFA has been indispensable for tracking aggregate progress of the field, but two challenges limit its use as a deployment decision-support tool. First, is known biases and over-annotations in the corpus of existing GO annotations [1, 5–8]. This rewards predictors for learning the annotation distribution rather than the underlying biology. Second, CAFA-style metrics operate on the bag-of-GO-terms projection of the prediction. They cannot directly assess whether a model’s biological rationale is mechanistically plausible, whether it has ignored organism context, or whether it has merely restated an existing domain-based inference.

The most systematic recent illustration of the gap between aggregate scoring and biological validity comes from de Crécy-Lagard *et al.* (2025) [9], who manually reviewed all 453 EC predictions made by DeepEC-Transformer — a top-performing deep-learning enzyme function predictor — for uncharacterised *E. coli* proteins. The result was striking: only 3/453 predictions were genuinely novel and correct. The remainder fell into reproducible, biologically explicable error classes involving paralog over-propagation, missing pathway context, in-vitro/in-vivo ambiguity, and frequency-biased repetition.

Each rejection in the de Crécy-Lagard analysis required the reviewer to **synthesise multiple lines of evidence**: domain architecture, paralog subfamily context, pathway presence/absence in the organism, genetic evidence, and orthogonal primary literature.

This is the **synthesis bottleneck**. Human experts can do it, but it becomes a serious review burden as prediction sets and method submissions grow. This motivates a scalable intermediate layer between aggregate metric evaluation and prospective experiment. We present **AI-AUGR (Assessment via Unified Gene-evidence Review)**, an evidence-grounded review framework that produces structured, human-readable assessments of existing and predicted annotations. AI-AUGR is designed as a *complement* to CAFA-style benchmarks rather than a replacement: it reviews biological rationales outside aggregate term metrics, surfaces systematic failure modes that metric aggregation hides, and distinguishes genuinely novel insight from restatement of existing domain-based annotation.

We make five contributions: (1) the AI-AUGR review workflow and schema for evidence-grounded evaluation of function-prediction outputs; (2) ESR-ECOLI-DET-Mini, a 7-gene Expert Synthetic Review recap of the de Crécy-Lagard *et al.* [9] *E. coli* error taxonomy as a positive control, plus an answer-key-withheld recapitulation that recovers the major error calls but not all expert boundary judgments; (3) two linked BioReason-Pro benchmarks: ARGO139, a manually selected 139-gene benchmark for RL-narrative review, and ARGO95, the 95-gene ARGO139 subset with HuggingFace-catalogue SFT term predictions; (4) a BioReason-Pro evaluation showing that SFT GO terms and RL narratives mostly restate existing pipelines while failing in systematic, biologically diagnosable ways; and (5) all benchmarks released as open resources at github.com/ai4curation/ai-gene-review.

2 Background and related work

2.1 Protein function prediction and its production pipelines

For production annotation, the key property of family- and homology-based pipelines is auditability. InterPro [10] first aggregates signatures from resources such as Pfam, SMART, PANTHER, CDD, ProSite, and TIGRFAM into protein families and domains. InterPro2GO then applies manually curated signature-to-GO mappings, so an electronic annotation can be traced from a sequence hit to a signature and then to a

specific curator-authored mapping. The PANTHER/PAINT pipeline [11] is similarly inspectable but phylogenetic: an expert curator reviews a family tree, annotates ancestral nodes with GO terms, and propagates those annotations to descendants via the IBA (Inferred from Biological ancestor) evidence code. Orthology-based pipelines (GOA electronic, Ensembl Compara, Reactome inference) operate through related transfer assumptions.

A key characteristic of these pipelines is their transparency and relative ease with which propagation errors can be corrected at source. A curator who disagrees with an annotation can usually identify the upstream signature, mapping, tree node, or orthology assertion that would need to change.

More recent methods have moved toward direct statistical prediction. Deep-learning predictors such as DeepGO(Plus) [12], DeepFRI, NetGO3, and PFMuL [13] consume sequence, structure, or network context and emit GO terms without the same explicit provenance trail. Protein language models (ESM1/2/3 [14], ProtT5, ProST5) provide unsupervised representations that outperform family-based features on several downstream tasks. A newer layer of reasoner-style systems, including GO-GPT and BioReason-Pro [15], adds generated rationales or functional summaries to predicted or upstream-provided term lists. These outputs are useful to read, but they make evaluation harder because their biological claims are only partly captured by bag-of-GO-terms metrics.

2.2 CAFA and the evaluation of function prediction

The Critical Assessment of Functional Annotation (CAFA) challenges [2–4] have provided the standard benchmark for protein function prediction for more than a decade. CAFA uses a temporal-holdout design: method developers submit predictions for a pre-specified protein set before a date; the community then waits for experimental annotations to accumulate in GOA; and predictions are scored against the resulting annotation targets. The primary metrics are $F_{\max}$ (maximum F-measure across decision thresholds) and $S_{\min}$ (minimum semantic distance, an information-content-weighted Resnik-style metric). Predictions are stratified by aspect (Molecular Function, Biological Process, Cellular Component) and by a “No-Knowledge” vs “Limited-Knowledge” category depending on whether the target protein had any prior annotation in the aspect.

CAFA has unambiguously shaped the field: performance has improved round over round, method development is empirically grounded, and the community has a shared vocabulary for comparing approaches. But a growing body of work identifies structural limitations in CAFA-style evaluation that the rise of reasoner-style predictors is making increasingly acute. We group these into three categories: metric-level, ground-truth, and holdout-set biases.

Metric-level limitations. $F_{\max}$ penalises specificity: a method that predicts only high-confidence, deep annotations for a small number of proteins is penalised in recall relative to one that blankets all proteins with frequent, generic terms [16]. $F_{\max}$ is also lenient toward false positives — over-prediction is less costly than under-prediction [16, 17]. Kahanda *et al.* [16] showed that switching between protein-centric and term-centric $F_{\max}$ reshuffles method rankings, meaning “which method is best” depends on the evaluation protocol. Clark and Radivojac [17] proposed an information-theoretic alternative ($S_{\min}$, based on misinformation and remaining uncertainty), which addresses some of these issues but still operates on GO-term sets rather than narrative content.

Ground-truth limitations. CAFA takes the GOA state as ground truth, but GOA is neither complete nor unbiased. Škunca *et al.* [6] found that less than 1% of experimental annotations are negative, making false-positive detection inherently difficult (the “open-world assumption”). Haynes *et al.* [7] showed that 58% of human GO annotations cover only 16% of genes, creating a self-reinforcing cycle in which well-

studied genes accumulate ever more annotations while poorly studied genes remain dark. Schnoes *et al.* [8] documented systematic biases in experimental annotations that distort our picture of protein function space. Tomczak *et al.* [18] demonstrated that GO enrichment results change with annotation version — the same experiment, re-analysed with a newer GOA release, yields different significant terms. For CAFA, this means the “correct answer” is a moving target: a prediction scored as a false positive in one round may become a true positive in the next.

Holdout-set biases. The temporal-holdout design means the benchmark set is determined by whichever proteins gain experimental annotations during the accumulation window. This is not random. GOA/EBI explicitly prioritises human disease-relevant proteins, proteins with no existing annotation, and proteins targeted by funded curation campaigns (e.g. Reference Genome Project, British Heart Foundation, Kidney Research UK grants) [19]. The CAFA3 paper [4] acknowledged that the prokaryotic benchmark was “heavily biased toward *Escherichia coli* K-12” and that “the process of acquiring GO annotations is highly biased, leading to a distribution of categories that changes over time.” The NK/LK (No-Knowledge / Limited-Knowledge) split reflects this: CAFA3 had only 147 NK-MF benchmarks, a small and skewed sample of whatever happened to get characterised during the window. A method evaluated against one CAFA holdout is being tested against a sample of biology that reflects the funding landscape and curation priorities of that particular year, not the distribution of biology that a database actually needs to annotate. CAFA3 partially addressed this by commissioning genome-wide experimental screens for *Candida albicans*, *Pseudomonas aeruginosa*, and *Drosophila melanogaster* [4], but these organism-specific screens introduce their own term-distribution biases tied to the phenotypes assayed.

Narrative and reasoning output require a different evaluation layer. For classical family-based predictors this limitation was academic: there was nothing outside the term list to evaluate. For reasoner-style predictors it is central. A prediction’s narrative summary, its cited domain evidence, its mechanistic reasoning, and its identification (or non-identification) of edge cases such as pseudoenzymes are all outside the bag-of-GO-terms projection used by $F_{\max}$ and $S_{\min}$.

Complementary evaluation strategies have been proposed — pathway-level benchmarks, molecular-function specificity scoring, organism-aware evaluation, curator-in-the-loop spot-checking — but none has been operationalised as a scalable, reusable pipeline. The de Crécy-Lagard *et al.* [9] study is the most systematic attempt to date, but it relies on exhaustive manual review, which does not scale with the rate at which new methods are released.

2.3 BioReason-Pro: a protein-function reasoner

BioReason-Pro [15] is a representative member of the reasoning-output generation of function predictors. Architecturally, it is a narrow fine-tune of the Qwen3-4B foundation LLM, exposed in two variants: **BioReason-Pro SFT**, trained on ~124K reasoning traces synthesised by GPT-5, and **BioReason-Pro RL**, further trained from the SFT checkpoint with GRPO on ~9.2K curated examples. BioReason-Pro does not consume sequence directly: its upstream predictor, **GO-GPT** (an autoregressive transformer trained on ESM2 embeddings and organism context), produces an initial GO term hierarchy, which BioReason-Pro then receives as input together with the InterPro domain list, known protein–protein interaction partners, and organism label. BioReason-Pro emits a <think> reasoning trace followed by a free-text functional summary; the downstream GO term list presented to the user in the web app is the GO-GPT input, not an independent BioReason-Pro prediction [15].

This architecture has two consequences for evaluation. The first is that much of the scientific content of a BioReason-Pro output is in the narrative, which requires direct biological review rather than term-overlap scoring alone. The second is that BioReason-Pro’s apparent skill is partially a function of how informative the

upstream InterPro domain labels happen to be for a given protein: when a family name is already diagnostic of function (e.g., “Armadillo repeat-containing”), the reasoning trace can recover a plausible summary; when it is not (e.g., “DUF1234”), the model must either confabulate or rely on correlations from the training distribution. Both modes are visible to an expert reading the output but are flattened by a GO-term-overlap score.

The BioReason-Pro paper itself highlights several impressive case studies, including correct *de novo* identification of SBP2 as an eEFSec binding partner (later validated by cryo-EM), correct pseudoenzyme identification for CFAP61, and biologically coherent predictions for synthetic anti-CRISPR proteins. These case studies are promising. But as case studies, they do not by themselves establish deployment performance across large protein sets. As we show below, pseudoenzyme identification in particular does not generalise to every broader test case.

2.4 Expert Synthetic Review precedent

The second central reference for this paper is de Crécy-Lagard *et al.* (2025) [9], who manually reviewed all 453 EC number predictions made by DeepECTransformer for *E. coli* proteins of unknown function. We use **Expert Synthetic Review** (ESR) for this role: an expert-authored biological-validity review that synthesises domain architecture, paralog context, pathway presence, genetics, and primary literature into prediction-level calls. The de Crécy-Lagard analysis is the source from which a future full ESR-ECOLI-DET benchmark can be extracted. In this manuscript we use a 7-gene quick-check subset, ESR-ECOLI-DET-Mini.

The headline finding from the full expert review — only 3/453 predictions were genuinely novel and correct — is already significant. For deployment, however, the more reusable contribution is the error taxonomy developed during that review (**Table 1**). It separates true novelty from training-data recapitulation, lower-specificity predictions, paralog-subfamily mistakes, organism-context mistakes, frequency-biased default labels, and cases where the available evidence does not justify a confident call.

Table 1. Expert Synthetic Review prediction-review taxonomy adapted from de Crécy-Lagard *et al.* [9] and used for AI-AUGR prediction assessment.

Code	Meaning	Biological question asked by the reviewer	Example pattern
COR	Correct novel prediction	Is the predicted activity supported and absent from existing curated annotation?	ygfF glucose 1-dehydrogenase, supported by SDR subgroup and biochemical evidence
CNN	Correct but not novel	Is the prediction correct but already present in the training or curated record?	Correct EC/GO calls already represented in existing annotations
LSP	Less precise	Is the prediction directionally correct but less specific than the existing annotation?	Broad family-level activity where a more specific substrate or reaction is known

Code	Meaning	Biological question asked by the reviewer	Example pattern
PLI	Paralog incorrect	Does the activity belong to a related but non-isofunctional paralog or subfamily?	yciO assigned TsaC-like TC-AMP synthase activity; yegV assigned KdgK-like substrate specificity
NPI	Non-paralog incorrect	Is the prediction refuted by pathway, genome, compartment, or organism context?	yjhQ assigned mycothiol synthase although <i>E. coli</i> lacks mycothiol biosynthesis; yrhB assigned QueD activity
REP	Repetition / frequency bias	Does the model default to a common training label despite incompatible sequence evidence?	fepE predicted as histidine kinase despite no histidine-kinase domain or similarity
UNC	Uncertain	Is the evidence insufficient to validate or reject the predicted function?	yjdM phosphonoacetate hydrolase: in-vitro activity without clear in-vivo role

The findings are useful because the same numerical verdict can arise for different biological reasons. yciO is a paralog problem: it is related to TsaC, but the measured TC-AMP production rate is orders of magnitude weaker and the protein does not complement the relevant in-vivo function. yjhQ and yrhB are pathway-context problems: the predicted reactions are assigned to proteins in a pathway or enzymatic role already absent or otherwise accounted for in *E. coli*. yjdM illustrates a boundary case, where an in-vitro activity is real but the physiological role remains unresolved. fepE is different again: it is a Wzz-family O-antigen chain-length regulator, so the histidine-kinase prediction is best understood as a high-frequency default rather than a plausible mechanistic hypothesis. This taxonomy provides both a motivation for the agentic-review approach and a positive-control target for checking whether AI-AUGR can encode the same expert error classes.

3 Methods

3.1 The AI-AUGR pipeline

AI-AUGR is an agentic curation pipeline in which an LLM curator-agent processes a per-gene evidence package and emits a structured review validated against a LinkML schema. The pipeline has four stages: evidence assembly, review, validation, and, when applicable, prediction scoring.

Evidence assembly. For each gene, the pipeline automatically collects (i) the UniProt flat-file record, (ii) the full GO annotation table for the gene from QuickGO, (iii) the InterPro domain architecture and family membership, (iv) cached publication records, with full text where available and abstract-level content otherwise, for publications cited in the GO annotation table, and (v) an orthogonal deep-research report generated

by a separate literature-retrieval agent with web access. The deep-research step is treated as a preliminary research summary by the reviewing agent, not as a primary source. All evidence is cached locally, so that reviews are reproducible and verifiable.

Curator-agent review. The curator-agent proceeds through three sequential phases. In the **annotation-level review** phase, the agent processes each existing GO annotation for the gene in turn, assigning a structured action from {ACCEPT, KEEP_AS_NON_CORE, MODIFY, REMOVE, MARK_AS_OVER_ANNOTATED, UNDECIDED} together with a supporting-text quote drawn verbatim from one of the cached publications. The quote must be literally present in the cache; this is enforced by the validator. In the **core-function synthesis** phase, the agent writes a free-text summary of the gene’s molecular function, biological role, and cellular location, proposes any missing GO terms, and optionally flags experimental or curatorial questions for a human expert. In the **prediction review** phase (used when evaluating an external predictor, including in the BioReason-Pro and de Crécy-Lagard studies below), the agent examines each annotation supplied by the external predictor, including terms already represented in GOA, and classifies it using the de Crécy-Lagard error taxonomy (COR / CNN / LSP / PLI / NPI / REP / UNC) together with a set of structured error-type tags (e.g., PARALOG_OVERANNOTATION, PATHWAY_CONTEXT_IGNORED, FREQUENCY_BIAS, IN_VITRO_NOT_IN_VIVO, TRAINING_DATA_CONTAMINATION).

Validation. All review outputs are validated against the LinkML GeneReview and PredictionReview classes. In addition to structural schema conformance, a custom validator enforces biological best-practice rules: every GO term identifier must exist in the current ontology, every supporting-text quote must appear literally in one of the cached publications for that gene, and every action with type MODIFY or REMOVE must carry a justification paragraph. Invalid reviews are returned to the agent for repair. The same validation rules are applied to individual reviews and to the full review corpus.

Implementation. AI-AUGR is implemented as a Python package (ai-gene-review). Gene reviews, the schema, the validator, and all evaluation data are released at github.com/ai4curation/ai-gene-review under an open licence. A public browser at <https://ai4curation.io/ai-gene-review/> renders individual reviews as HTML for human inspection.

3.2 ARGO139 construction and BioReason-Pro evaluation

For the BioReason-Pro analyses we use two linked denominators. **ARGO139** (Annotation Review GO) contains 139 proteins selected manually from genes for which we could assemble an AI-AUGR evidence package and a BioReason-Pro RL output. **ARGO95** is the 95-gene ARGO139 subset with public HuggingFace-catalogue BioReason-Pro SFT GO-term predictions. The selection was not intended to approximate a random UniProt sample. Instead, it was designed to test deployment-relevant biology: well-characterised model-organism proteins; species with specialist organism resources; species without comparable MOD-backed coverage, especially *Bacillus subtilis*; and edge cases such as pseudoenzymes, paralog families, sporulation sigma factors, organism-specific regulators, moonlighting proteins, a phage protein, and a snake-venom metalloproteinase.

ARGO139 spans 14 species labels. Most genes (115/139) come from species with model-organism or specialist curation resources; 24/139 come from non-MOD or less-specialized contexts, including *B. subtilis*, *Pseudomonas putida*, a phage entry, a southern copperhead venom enzyme, and a mosquito protein. The full member list is `projects/BIOREASON_COMPARISON/genes.csv`; derived composition files are `argo139-species-counts.csv` and `argo139-curation-context-counts.csv`.

Table 2. ARGO139 species composition.

Species label	Display label	Genes	Curation context
SCHPO	<i>S. pombe</i>	23	Model organism / specialist curation resource
human	<i>H. sapiens</i>	19	Model organism / specialist curation resource
worm	<i>C. elegans</i>	15	Model organism / specialist curation resource
BACSU	<i>B. subtilis</i>	13	Non-MOD or less-specialized species
ECOLI	<i>E. coli</i>	13	Model organism / specialist curation resource
rat	<i>R. norvegicus</i>	12	Model organism / specialist curation resource
mouse	<i>M. musculus</i>	11	Model organism / specialist curation resource
yeast	<i>S. cerevisiae</i>	11	Model organism / specialist curation resource
DROME	<i>D. melanogaster</i>	8	Model organism / specialist curation resource
PSEPK	<i>P. putida</i>	8	Non-MOD or less-specialized species
ARATH	<i>A. thaliana</i>	3	Model organism / specialist curation resource
9CAUD	9CAUD phage	1	Non-MOD or less-specialized species
AGKCO	<i>A. contortrix</i>	1	Non-MOD or less-specialized species
ANOGA	<i>A. gambiae</i>	1	Non-MOD or less-specialized species

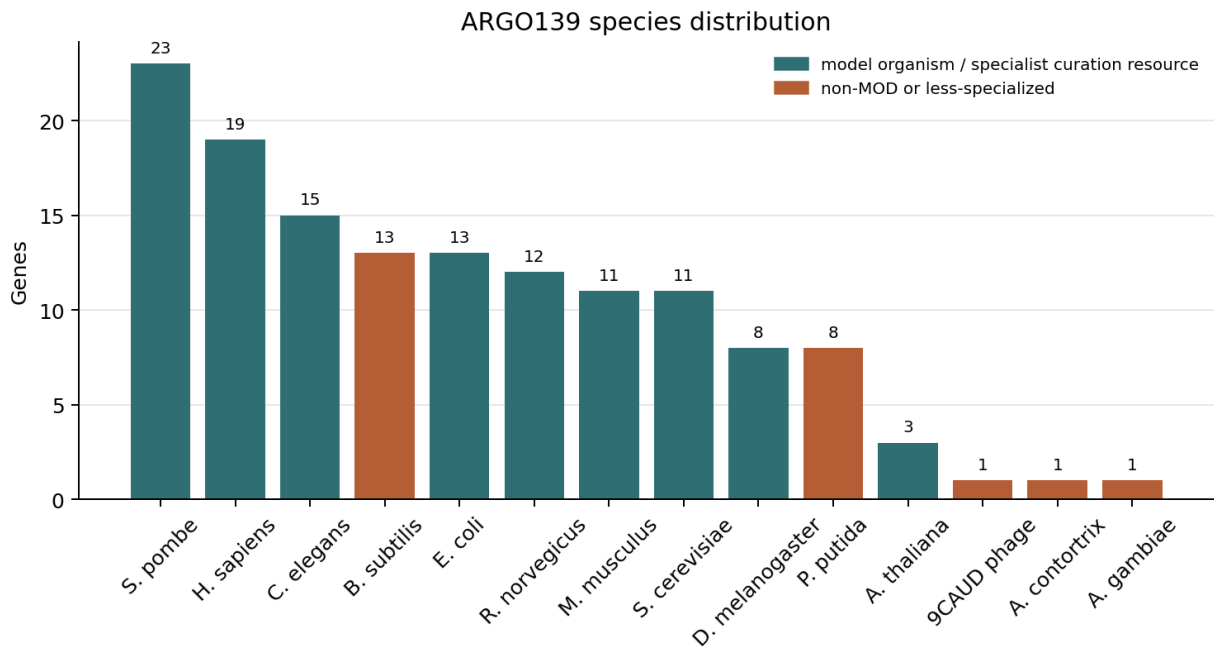


Figure 1: ARGO139 species distribution. Teal bars mark model-organism or specialist-resource species; orange bars mark non-MOD or less-specialized species.

For each ARGO139 gene we obtained the BioReason-Pro RL functional summary and reasoning trace from the public BioReason-Pro web application (app.bioreason.net). We also ran the AI-AUGR pipeline to produce an agent-adjudicated local reference review. These references were not independently expert-signed: as of the current refresh, 73/139 were COMPLETE, 46 DRAFT, 16 IN_PROGRESS, and 4 INITIALIZED. A dedicated **comparison agent** then scored each BioReason-Pro RL output against the AI-AUGR review along two axes, each on a 1–5 scale:

- **Correctness:** 5 = all material claims supported; 4 = core function correct with one limited secondary

error; 3 = core function correct but one or more substantive claims wrong; 2 = some relevant biology but errors materially distort the function; 1 = fundamental mischaracterisation.

- **Completeness:** 5 = covers core functions, locations, complexes, and pathway context; 4 = core plus most major context with a limited omission; 3 = gets the basics but misses significant biology; 2 = only one important facet or substantially incomplete; 1 = superficial or misses the core function.

Omissions were charged to completeness rather than correctness. The comparison agent wrote an evidence-based rationale and a qualitative comparison against the supplied InterPro domain/family labels: does BioReason-Pro add biological insight beyond those labels, or narratively restate them? Only 92/139 genes had a GO_REF:0000002 annotation in the committed GOA snapshot, so we treat actual InterPro2GO as a baseline only for that subset. Reviews were stored per gene as {GENE}-bioreason-rl-review.md, with the raw BioReason-Pro export cached as {GENE}-bioreason-rl-predictions.md.

ARGO139 is the collected export cohort, not the unconditional performance denominator. We excluded **csr-1** from model-performance aggregates because the supplied record and sequence were nhr-47/Q17370 rather than CSR-1. Seven retained exports were exact 2,000-residue prefixes of longer UniProt sequences (Dscam1, BRCA2, HTT, LRRK2, human and mouse NOTCH1, and TOR1) and were flagged as sequence-truncated. The frozen policy and per-gene manifest record export timestamps, expected and cached accessions, sequence lengths, reference status, latest GOA annotation date, and SHA-256 checksums.

To measure score reproducibility, a second rater independently scored a deterministic 20-gene subset. We balanced the sample by taking four genes from each first-rater correctness stratum; within each stratum selection was the SHA-256 order of a fixed release salt plus species/gene key. The second rater could read only the raw RL export and local AIGR reference, not the first review, project metrics, issue, or git history. We report exact and within-one agreement, mean absolute error, Pearson and Spearman association, and quadratic-weighted Cohen’s kappa. This balanced design tests the full rubric range rather than estimating prevalence-weighted ARGO139 agreement.

For SFT GO-term review we used only the public HuggingFace wanglab/protein_catalogue source. This defines ARGO95: the 95/139 ARGO139 genes present in that snapshot, with 955 SFT GO-term predictions. We did not fill the remaining 44 ARGO139 genes from the BioReason-Pro web export in the primary SFT analysis because that export exposes a fundamentally different high-recall GO-term panel, including many ancestor and obsolete terms. Web-export SFT terms, the SFT narrative cross-check, and a separate GO-GPT overlap review are retained only as supplemental material.

We also rebuilt the ARGO139 GO-GPT leaf files from the raw web exports. Ontology-aware pruning retained 5,923 terms: 1,900 exact positive AIGR matches (CNN), 129 exact rejected or over-annotated matches (NPI), and 3,894 unresolved terms (UNC). Documents with unresolved placeholders are marked DRAFT; 138/139 therefore remain pending, and this set is not reported as a completed GO-GPT performance evaluation.

3.3 Case study 2: ESR-ECOLI-DET-Mini recap

From the 453 DeepECTransformer predictions in de Crécy-Lagard *et al.* [9] we selected 7 genes spanning all error classes represented in the paper, specifically: **ygff** (COR), **yciO** and **yegV** (PLI), **yjhQ** and **yrhB** (NPI), **yjdM** (UNC), and **fepE** (REP). This quick-check benchmark is ESR-ECOLI-DET-Mini (alias ESR-ECOLI-DET-7; Zenodo dataset [20]). Selection was deterministic from the published tables and used the published classes. The current repository artifacts are therefore not blinded: the project file lists the paper’s verdicts, the gene reviews can cite PMID:40703034, and the structured prediction-review files quote the de Crécy-Lagard rationale directly. We treat this exercise as a positive-control reproduction of the tax-

onomy, not as an independent validation of AI-AUGR accuracy.

For each gene we ran the full AI-AUGR pipeline to produce a gene review, and separately produced a `predictions-review.yaml` for the DeepECTransformer prediction itself, using the same COR, CNN, LSP, PLI, NPI, REP, and UNC taxonomy and structured error-type tags described in §3.1. After review completion we compared the agent’s classification and rationale against the published classification in [9].

To probe how much of this result depends on label/rationale leakage, we also archived an independent answer-key-withheld recapitulation experiment. This run was performed on an ablated branch in which review content was regenerated, and the reviewer reported that the de Crécy-Lagard source paper, including PMID:40703034 and the published rationales, was not used. The evidence package was not restricted to curated annotations alone: it used UniProt, GOA, the DeepECTF paper, primary literature, and one in-house bioinformatics analysis for *yciO*. We therefore treat it as a literature- and bioinformatics-assisted Expert Synthetic Review recap with the answer key withheld, not as a strict curated-annotation-only benchmark. The result bundle is preserved under `projects/BIOREASON_COMPARISON/recapitulation-experiment/claude-expt-1/`; all 7 copied gene reviews and all 7 copied prediction reviews validate against the LinkML schema.

3.4 Reproducibility

All reviews, raw predictions, local AIGR references, and validator configurations for both case studies are in the repository under `genes/{ORG}/{GENE}/` and `projects/BIOREASON_COMPARISON/`. `genes.csv` is the ARGO139 member list; `argo139-species-counts.csv`, `argo139-curation-context-counts.csv`, `benchmark-cohorts.csv`, and `benchmark-genes.csv` enumerate the cohort composition and source provenance. The GO hierarchy and ID-label audit use the official archived 2026-03-25 `go-basic.obo`; its SHA-256 is pinned in the benchmark policy and release-specific active and obsolete sentinel terms are checked at load time. GOA snapshots may retain identifiers after ontology obsolescence. An independent verification script downloads the archive and queries QuickGO and OLS; on 2026-07-12 its checksum matched and both live services confirmed the disputed obsolete sentinels. Exact frozen-GOA matches are normally CNN, and ontology status is reported separately from biological assessment. A status-only mismatch retains its CNN, COR, or UNC call; LSP remains reserved for a canonical concept that is more generic than the supported annotation. The ESR-ECOLI-DET-Mini result bundle is archived on Zenodo [20]. Supplemental source-availability details are in [supplemental-benchmark-details.md](#). The repository contains the scripts and configuration needed to reproduce each stage. The full corpus of 139 BioReason-Pro reviews and 7 *E. coli* prediction reviews is also browsable at the public site.

4 Results

4.1 AI-AUGR recapitulates major expert failure calls but not all nuance

We first asked whether AI-AUGR can represent and apply an existing expert biological-validity taxonomy before using it to evaluate BioReason-Pro. In the original ESR-ECOLI-DET-Mini artifacts, all 7 AI-AUGR prediction-review classifications matched the published de Crécy-Lagard classifications and recorded substantially the same mechanistic rationale (**Table 3**). Because the paper’s labels and rationales were available in those artifacts, this result is a positive control rather than a blinded benchmark.

The answer-key-withheld recapitulation gives a more realistic estimate of what an agentic synthetic reviewer can do. It recovered 4/7 exact labels: *fepE* (REP), *yciO* (PLI), *yjhQ* (NPI), and *yrhB* (NPI). The misses were biologically informative rather than random. For *yegV* and *ygff*, the reviewer chose UNC rather than the

expert PLI/COR calls, reflecting conservative uncertainty around substrate identity and in-vivo role. For yjdM, the reviewer called NPI rather than UNC, over-converting an in-vitro-versus-in-vivo ambiguity into a rejection. Thus the agent did not reach the same level of expert nuance, but it did identify several high-value smell-test failures: frequency-biased histidine-kinase overprediction, paralog overannotation, and pathway-context violations.

Table 3. Positive-control reproduction and answer-key-withheld recapitulation of de Crécy-Lagard *et al.* [9] on 7 *E. coli* DeepECTransformer predictions.

Gene	Paper class	Positive-control AI-AUGR class	Answer-key- withheld class	Interpretation
ygfF	COR	COR	UNC	Conservative miss; in-vitro SDR/glucose-DH evidence judged insufficient for in-vivo function
yciO	PLI	PLI	PLI	Paralog-overannotation call recovered
yegV	PLI	PLI	UNC	Conservative miss around sugar-kinase substrate identity
yjhQ	NPI	NPI	NPI	Pathway-absence call recovered
yrhB	NPI	NPI	NPI	Wrong-enzyme/pathway-context call recovered
yjdM	UNC	UNC	NPI	Overcalled an in-vitro/in-vivo ambiguity as incorrect
fepE	REP	REP	REP	Frequency-bias histidine-kinase call recovered

The recorded rationales are important, not just the class labels. In the yciO case, both runs converged on a paralog-subfamily argument. In yjhQ and yrhB, the withheld run recovered organism/pathway-context failures. In fepE, it recognised the Wzz O-antigen length-regulator assignment and the absence of a histidine-kinase domain. These are precisely the kinds of triage signals a database team would want before spending expert time on a large sequence-AI prediction set. The disagreements also set the boundary: agentic review is useful for prioritising suspicious predictions and detecting recurring error modes, but current outputs should still be treated as curator-assistive rather than curator-equivalent.

4.2 ARGO95 SFT GO-term review

We next reviewed BioReason-Pro SFT GO-term predictions on ARGO95. Each predicted term was classified against GOA and the AI-AUGR review using the Expert Synthetic Review taxonomy: terms already in GOA were assigned CNN; terms absent from GOA but supported by AI-AUGR core functions or accepted annotations were reviewed as COR candidates; terms contradicted by the AI-AUGR review were reviewed as NPI candidates; less precise terms were assigned LSP; high-frequency generic terms were assigned REP; and terms not supported or refuted by available evidence were assigned UNC.

ARGO95 uses the cleaner HuggingFace wanglab/protein_catalogue source: 95 ARGO139 genes and 955 terms. We exclude the web-export-only SFT terms from the primary analysis because they expose a much larger GO ancestor hierarchy and are not comparable to the HF catalogue term set. The web-export view remains in the supplement as a source-diagnostic check, not as part of the main SFT denominator.

Table 4. ARGO95 SFT GO-term assessment distribution.

Genes	Terms	CNN	NPI	PLI	COR	LSP	REP	UNC
95	955	679 (71.1%)	117 (12.3%)	5 (0.5%)	24 (2.5%)	44 (4.6%)	29 (3.0%)	57 (6.0%)

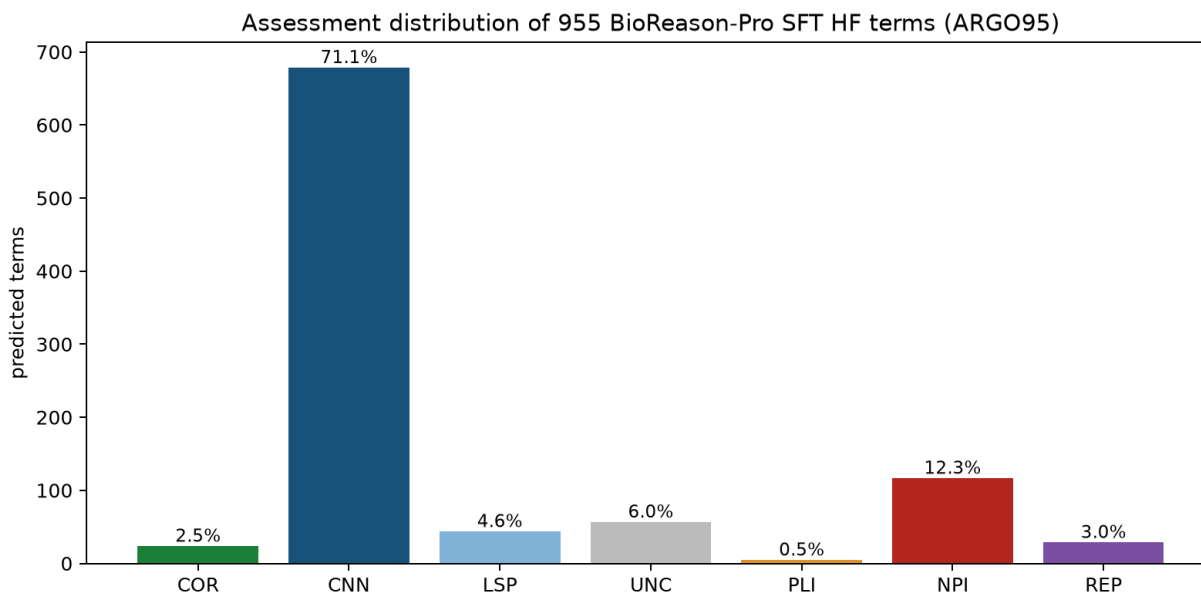


Figure 2: Assessment distribution for 955 BioReason-Pro SFT HF-catalogue terms on ARGO95. CNN dominates; COR terms are known biology absent from GOA, not functions unknown to the literature.

ARGO95 gives the cleanest estimate of SFT term quality: 679/955 (71.1%) of SFT terms are correct but non-novel, including 632 exact frozen-GOA matches; 151/955 (15.8%) are NPI, PLI, or REP; and only 24/955 (2.5%) are correct terms absent from frozen GOA that a database might consider importing. The COR calls are not discoveries of biology unknown to the literature. They are known functions missing from GOA or captured less precisely than the model's term, such as RAS2 activation of adenylate cyclase, Atg2 inter-membrane lipid transfer, regulation of mitochondrial gene expression by atfs-1, and the phosphatidylinositol deacylase activity of bst1.

As a Tier-1 baseline, we also computed a retrospective CAFA-style GOA-agreement score for the same SFT terms. Because BioReason-Pro SFT terms are unranked, this is a single-threshold F_1 rather than $F_{\max}$. After propagating predictions and current GOA annotations over GO `is_a` and `part_of` ancestors, the HF subset scored precision 0.865, recall 0.479, and $F_1 = 0.617$ against current GOA; exact-term F_1 was 0.418. This makes the prediction set look reasonably concordant with GOA, but the biological review shows why agreement is not enough: 53/152 HF terms labelled NPI, PLI, or REP were exact matches to current GOA, and 124/152 had propagated overlap with current GOA. A CAFA-style score can therefore be a useful Tier-1 baseline, but it cannot distinguish restatement from novelty or current database agreement from biological validity.

The excluded web-export subset makes a different point. Because it contains many ancestor terms, even leaf filtering leaves a much larger and more uncertain term panel than the HF catalogue. We therefore treat web-export SFT terms as a supplemental source-availability diagnostic, not as interchangeable observations in the ARGO95 SFT analysis.

The incorrect SFT terms recapitulate the same biological failure modes seen in the RL narratives: pseudoenzyme overcalls, holdase/foldase confusion in bacterial chaperones, wrong compartments, antibiotic-target/response confusion, and substrate/partner role reversal. The RAS2 and CpxP cases also show that the term and narrative arms can fail independently. RAS2’s SFT GO terms include the core adenylate-cyclase activation biology that the RL narrative missed; CpxP’s SFT terms place it in the periplasm while the RL narrative calls it cytoplasmic. For database deployment, neither output should be trusted without reviewing the other.

4.3 ARGO139 RL narrative review

Across the ARGO139 performance set (138/139 collected exports after the wrong-input exclusion), BioReason-Pro RL achieved a mean correctness of **4.0/5** and a mean completeness of **2.9/5** against the agent-adjudicated local AIGR reference. The two axes are reported separately; their empirical association was Pearson 0.688 and Spearman 0.617, rather than the weak association asserted in the earlier analysis. The score distribution is shown in **Table 5**.

Table 5. Score distribution for BioReason-Pro RL on the 138-gene ARGO139 performance set.

Score	Correctness	Completeness
5	70 (51%)	1 (1%)
4	25 (18%)	40 (29%)
3	22 (16%)	51 (37%)
2	14 (10%)	39 (28%)
1	7 (5%)	7 (5%)

The high-correctness tail is real: 70 genes scored 5/5 on correctness. But completeness is much weaker; only one gene scored 5/5 on both axes (Uggt1, whose InterPro family name is itself highly diagnostic).

In the blinded 20-gene second review, correctness agreement was 80% exact and 100% within one point (mean absolute error 0.20; quadratic-weighted kappa 0.950). Completeness agreement was 55% exact and 95% within one point (mean absolute error 0.50; kappa 0.744). Both axes matched exactly for 55% of cases. The lower completeness agreement identifies coverage scoring as the less reproducible judgment and supports reporting calibration data alongside the aggregate means. Mouse has the highest selected-case descriptive correctness mean (4.7), followed by *B. subtilis* (4.5), rat (4.4), and human (4.3), whereas the

selected *S. pombe* cases have the lowest mean (3.2). This comparison is confounded by deliberate case mix: the *S. pombe* set is enriched for obscure proteins and pseudoenzymes, while the mammalian set contains more canonical proteins. Training-distribution skew and InterPro-label informativeness remain hypotheses rather than measured explanations.

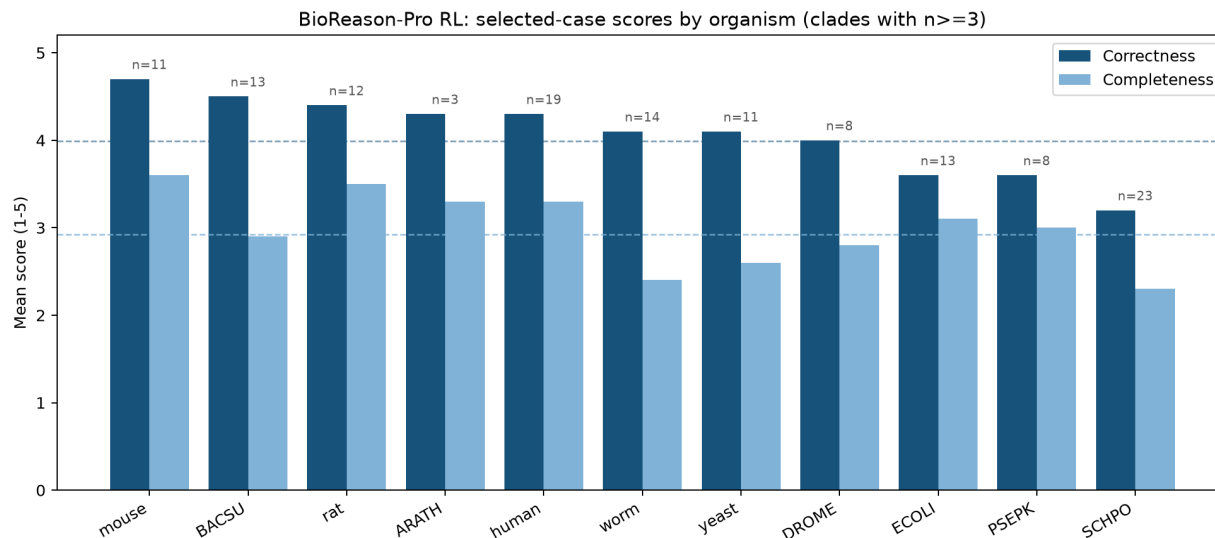


Figure 3: Descriptive per-organism correctness and completeness for the selected BioReason-Pro RL cases. Dashed lines show overall means and sample size is shown above each bar pair. Uneven case selection precludes causal or population-level interpretation.

The failures are not random. We identified eight reproducible model-output modes in the RL narratives (**Table 6**). Recording them as error modes is immediately useful for curation triage, because each points to a specific missing capability: catalytic-residue checking, signal-peptide and transmembrane reasoning, paralog-subfamily discrimination, organism-context reasoning, and detection of training-set leakage.

Table 6. Representative ARGO139 RL narrative failure modes.

Failure mode	Gene (organism)	Brief description of the failure
Pseudoenzyme blind spot	Epe1 (<i>S. pombe</i>)	Claims JmjC demethylase activity despite degenerate active site and no in-vitro activity
Pseudoenzyme blind spot	cts2 (<i>S. pombe</i>)	Called active chitinase despite lacking catalytic glutamate
Localisation default	CpxP (<i>E. coli</i>)	Called cytoplasmic; is periplasmic with a cleavable signal peptide
Localisation default	ETR1 (<i>A. thaliana</i>)	Called soluble cytoplasmic signal transducer; is an ER-membrane receptor
Paralog indistinguishability	Fyn / Src (mouse)	Generic Src-family kinase descriptions; misses paralog-specific biology

Failure mode	Gene (organism)	Brief description of the failure
Paralog indistinguishability	sigF/sigG/sigK (<i>B. subtilis</i>)	Treated as generic sigma factors; compartment-stage sporulation biology absent
Organism-specific biology absent	daf-16 (<i>C. elegans</i>)	Generic FoxO; no IIS/longevity/dauer biology
Organism-specific biology absent	atfs-1 (<i>C. elegans</i>)	Generic bZIP; misses UPRmt master regulator identity
Neo-functionalisation missed	Nmnat (<i>Drosophila</i>)	NAD biosynthesis enzyme; moonlighting neuroprotection role missed
Narrative/GO disconnect	RidA (<i>E. coli</i>)	Narrative mechanism mostly right; emitted term is generic protein binding
Cross-kingdom fold bias	aprE (<i>B. subtilis</i>)	Subtilisin described with human hemostasis and blood-coagulation processes
Cross-kingdom fold bias	PGRPLB (<i>A. gambiae</i>)	Mosquito protein described as a fruit-fly protein
Generated UniProt-style fabrication	Slc5a1 (rat)	Model-generated “UniProt Summary” claims steroid-sulfate transport; the cached UniProt record describes glucose/galactose cotransport

The comparison with supplied InterPro domain and family labels explains why the aggregate scores are misleading. Across ARGO139 the dominant mode is narrative restatement: BioReason-Pro translates those labels into prose without adding new biology. It succeeds when domain architecture is diagnostic (TOR1, NOTCH1, PTEN, EGFR, spo0A) and fails when the biology depends on catalytic-site loss, lineage-specific context, paralog identity, or compartmental signals not encoded in the family name. For the 92/139 genes with an actual InterPro2GO annotation, that mapping provides a separate baseline; where a mapping or supplied family label is wrong or uninformative, BioReason-Pro usually repeats the problem rather than repairing it.

5 Discussion

5.1 What agentic review adds

Agentic review surfaces information absent from aggregate scores, and this information is what annotation database leads need for deployment decisions.

First, it can read the narrative. BioReason-Pro emits free-text summaries and reasoning traces whose scientific content cannot be projected onto a bag of GO terms. AI-AUGR can grade those claims directly against cached literature and existing annotation.

Second, it surfaces reproducible failure modes. In the ARGO139 performance set, the RL mean correct-

ness of 4.0/5 hides failures that are biologically patterned: pseudoenzyme overcalls, localisation defaults, paralog indistinguishability, missing organism-specific biology, missed neo-functionalisation, narrative/GO disagreement, and cross-kingdom bias. These are not merely lower scores; they are deployment warnings.

Third, it distinguishes novelty from restatement. The SFT GO-term review shows that most terms are CNN, not new biology. The COR terms are useful curation candidates, but they are known functions absent from GOA, not discoveries unknown to the literature. A CAFA-style agreement score cannot make that distinction.

5.2 Why ARGO139

The earlier draft separated the RL narrative, SFT term, and GO-GPT overlap analyses into different denominators. That made the story unnecessarily fragile. The current analysis uses ARGO139 as the fixed biological gene set for RL narrative review and ARGO95 as the HF-only SFT term subset nested inside ARGO139.

ARGO139 is still not a population sample. It was deliberately constructed to span characterized biology, model-organism resources, and non-MOD or less-specialized contexts such as *B. subtilis*. That design is appropriate for deployment triage because it enriches for the edge cases where production databases most need review: pseudoenzymes, paralogs, lineage-specific biology, compartments, and moonlighting proteins. It should not be interpreted as an estimate of BioReason-Pro performance on a random UniProt draw.

The main remaining complexity is source availability. The HuggingFace SFT snapshot contains 95/139 ARGO139 genes; the remaining 44 can only be filled from a web export that exposes a different, ancestor-rich term panel. We therefore use ARGO95 for primary SFT term review and keep the larger source-availability views and GO-GPT overlap review in the supplement rather than in the main denominator story.

5.3 ESR as a positive control and recapitulation stress test

A natural concern is that an LLM curator-agent might share failure modes with the predictor it evaluates. The ESR-ECOLI-DET-Mini positive-control exercise does not eliminate that concern, because it was not blinded. Its value is narrower: it shows that AI-AUGR can encode the same expert taxonomy and assemble the same kinds of mechanistic rationale for pathway-presence reasoning, paralog-subfamily reasoning, in-vitro/in-vivo distinction, and training-label-frequency-bias detection.

The answer-key-withheld recapitulation narrows this concern but does not remove it. Recovering 4/7 exact labels is not expert equivalence; the agent was conservative on *ygfF* and *yegV* and too severe on *yjdM*. But the run recovered the most operationally important incorrect-call patterns. For deployment triage, this is the relevant signal. A database team does not need an agent to be a final arbiter to benefit from it; it needs the agent to flag suspicious sequence-AI outputs, identify the likely failure mode, and route the case to a human curator with a traceable rationale. A true external-validity test still requires a fresh blinded set whose labels and rationales are withheld from both the evidence package and the reviewing agent.

5.4 Three tiers of function-prediction evaluation

It is useful to separate function-prediction evaluation into three tiers.

Tier 1: aggregate metric evaluation. CAFA-style $F_{\max}$ and $S_{\min}$ scale to large holdouts and enable method comparison, but they inherit GOA-as-ground-truth bias, metric bias toward generic terms, and do not directly grade narratives.

Tier 2: expert or agentic biological-validity review. A domain expert, or an LLM curator-agent grounded in primary evidence, reads the prediction and classifies it using a biological taxonomy such as

COR/CNN/LSP/NPI/REP/UNC. This is the tier AI-AUGR partially automates. It is the right level for deployment decisions.

Tier 3: prospective experimental validation. This is the strongest evidence but the least scalable. Protein function is multi-dimensional: validating MF, BP, and CC claims can require different assays, organisms, and conditions. Tier 3 is essential for discovery claims, but not a practical first-pass filter for every computational prediction.

5.5 What we learned about BioReason-Pro

BioReason-Pro is not ready for unsupervised import into a production annotation database. On ARGO95, its SFT terms mostly restate GOA, with a non-trivial incorrect fraction in the cleaner HF subset. Its RL narratives are often plausible prose versions of supplied InterPro domain and family labels, not independent biological reasoning. The model is useful when domain architecture is diagnostic, but fails when the answer depends on catalytic-site loss, organism context, paralog identity, or compartmental signals.

The model’s best contribution may be format rather than current accuracy. A narrative can be reviewed, corrected, and triaged in ways that a naked GO-term list cannot. But the narrative and term arms must be reviewed independently, because they can disagree.

6 Limitations

Agent bias. AI-AUGR itself depends on an LLM curator-agent. We mitigate this through cached evidence, literal-quote checking, and schema validation, but shared blind spots remain possible. The current ESR-ECOLI-DET-Mini reproduction is useful as a positive control, not as a blinded external validation. The answer-key-withheld recapitulation is more informative, but its 4/7 exact-label result shows that the agent still lacks expert-level boundary judgment.

ARGO139 sample selection. ARGO139 is manually selected for characterized biology and hard deployment cases. It is not random, and its estimates should not be read as population-level BioReason-Pro performance.

SFT subset availability. ARGO95 covers the 95/139 ARGO139 genes available in the HF catalogue. The remaining 44 genes can be represented only by web-export SFT terms, but those terms include ancestor and obsolete-term inflation and are therefore treated as a supplemental source diagnostic rather than primary SFT observations.

Rubric subjectivity. The 1-5 Correctness and Completeness scales involve judgement. We mitigate this by requiring supporting evidence and releasing the per-gene reviews for inspection. A blinded 20-gene second review found higher reproducibility for correctness (quadratic-weighted kappa 0.950) than completeness (0.744). A larger, prevalence-weighted multi-rater study remains future work.

Single predictor in depth. We evaluate BioReason-Pro in detail, plus DeepECTransformer through the ESR-ECOLI-DET-Mini positive-control reproduction. Comparative evaluation across multiple reasoner-style systems is out of scope.

Label leakage and small recapitulation set. The original 7-gene ESR-ECOLI-DET-Mini exercise was selected from the published tables and the repository artifacts include the published labels and rationales. It should not be described as blinded. The archived answer-key-withheld recapitulation removes the most direct leakage source, but it is still small, selected after the de Crécy-Lagard study was known, and literature/bioinformatics-assisted rather than a locked curated-annotation-only benchmark. A stronger

follow-up would hold out the expert labels, remove PMID:40703034 from the per-gene evidence package, pre-register the evidence mode, and score the agent’s classifications before unblinding.

Reference-review staleness. The agent-adjudicated local AIGR references can become stale as literature and curation standards evolve. The pipeline is reproducible, and the browse site exposes generated review files for future reruns.

7 Conclusions

AI-AUGR reproduces the Expert Synthetic Review taxonomy on the 7-gene ESR-ECOLI-DET-Mini positive-control set, showing that the schema and review workflow can encode biologically meaningful error classes. In an answer-key-withheld recapitulation, it recovered 4/7 exact expert labels and the main high-value incorrect-call patterns, but not all expert boundary judgments. Applying AI-AUGR to ARGO139 shows that BioReason-Pro mostly restates existing annotation pipelines, occasionally names known biology missing from GOA, and makes systematic errors that aggregate metrics would hide.

The practical conclusion is not that CAFA-style metrics should be abandoned. It is that deployment decisions need a Tier 2 layer: expert or agentic review that can read narratives, classify novelty, and identify biologically meaningful failure modes. Agentic review is not yet a human-curator replacement, but it is already useful as a scalable smell test for sequence-AI predictions. ARGO139 and ARGO95 are our first linked benchmarks for that layer across BioReason-Pro RL narratives and SFT terms.

8 Figures

Figures are embedded inline. Additional planned figures for final submission:

- **AI-AUGR pipeline diagram.** Evidence assembly (UniProt, GOA, InterPro, cached literature, deep-research report) → curator-agent three-phase review → LinkML validation → structured output. (*To be created as a schematic.*)
- **CAFA evaluation context.** For representative CAFA $F_{\max}$ curves illustrating the aggregate evaluation paradigm, see Figure 2 of Radivojac *et al.* [2] and Figure 1 of Zhou *et al.* [4]. Our work addresses the gap between such aggregate curves and the per-protein biological validity that deployment decisions require.

9 Data and code availability

Repository: github.com/ai4curation/ai-gene-review — ARGO139 BioReason-Pro RL narrative reviews and ARGO95 SFT GO-term reviews, [article/supplemental-benchmark-details.md](#), ESR-ECOLI-DET-Mini 7-gene *E. coli* positive-control reproduction and archived answer-key-withheld recapitulation results in [projects/BIOREASON_COMPARISON/recapitulation-experiment/claude-expt-1/](#), archived on Zenodo [20], AI-AUGR pipeline, LinkML schema, and validator.

Browsable reviews: <https://ai4curation.io/ai-gene-review/>

References

- [1] The Gene Ontology Consortium. The gene ontology knowledgebase in 2023. *Genetics*, 224:iyad031, 2023.

- [2] Predrag Radivojac et al. A large-scale evaluation of computational protein function prediction. *Nature Methods*, 10:221–227, 2013.
- [3] Yuxiang Jiang et al. An expanded evaluation of protein function prediction methods shows an improvement in accuracy. *Genome Biology*, 17:184, 2016.
- [4] Naihui Zhou et al. The cafa challenge reports improved protein function prediction and new functional annotations for hundreds of genes through experimental screens. *Genome Biology*, 20:244, 2019.
- [5] Pascale Gaudet and Christophe Dessimoz. Gene ontology: pitfalls, biases, and remedies. In *The Gene Ontology Handbook*, volume 1446 of *Methods in Molecular Biology*, pages 189–205. 2017.
- [6] Nives Škunca, Adrian Altenhoff, and Christophe Dessimoz. Quality of computationally inferred gene ontology annotations. *PLoS Computational Biology*, 8:e1002533, 2012.
- [7] Winston A. Haynes et al. Gene annotation bias impedes biomedical research. *Scientific Reports*, 8:1362, 2018.
- [8] Alexandra M. Schnoes et al. Biases in the experimental annotations of protein function and their effect on our understanding of protein function space. *PLoS Computational Biology*, 9:e1003063, 2013.
- [9] Valérie de Crécy-Lagard et al. Limitations of current machine learning models in predicting enzymatic functions for uncharacterized proteins. *G3*, 15(10):jkaf169, 2025. PMID: 40703034.
- [10] Typhaine Paysan-Lafosse et al. Interpro in 2022. *Nucleic Acids Research*, 51:D418–D427, 2023.
- [11] Pascale Gaudet, Michael S. Livstone, Suzanna E. Lewis, and Paul D. Thomas. Phylogenetic-based propagation of functional annotations within the gene ontology consortium. *Briefings in Bioinformatics*, 12:449–462, 2011.
- [12] Maxat Kulmanov and Robert Hoehndorf. Deepgoplus: improved protein function prediction from sequence. *Bioinformatics*, 36:422–429, 2020.
- [13] Qiang Yuan et al. Structure-aware protein function prediction using graph convolutional networks. *Nature Communications*, 14:1234, 2023.
- [14] Zeming Lin et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379:1123–1130, 2023.
- [15] Alireza Fallahpour et al. Bioreason-pro: Advancing protein function prediction with multimodal biological reasoning. *bioRxiv*, 2026. doi: 10.64898/2026.03.19.712954.
- [16] Indika Kahanda, Christopher Funk, Fahad Ullah, Karin Verspoor, and Asa Ben-Hur. A close look at protein function prediction evaluation protocols. *GigaScience*, 4:s13742–015–0082–5, 2015.
- [17] Wyatt T. Clark and Predrag Radivojac. Information-theoretic evaluation of predicted ontological annotations. *Bioinformatics*, 29:i53–i61, 2013.
- [18] Aleksandra Tomczak et al. Interpretation of biological experiments changes with evolution of the gene ontology and its annotations. *Scientific Reports*, 8:5115, 2018.
- [19] UniProt-GOA. Uniprot-goa curation priorities. <https://www.ebi.ac.uk/GOA/newto>, 2026. Accessed April 2026.
- [20] Christopher J. Mungall. Independent clade reviews of 7 e. coli genes and deepctf prediction assessments (esr-ecoli-det-mini, clade-expt-1). <https://doi.org/10.5281/zenodo.20751016>, June 2026. Dataset.